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END SEMESTER EXAMINATION 2025

Name of the Course: B.Tech Petroleum Engineering

Semester: V

Name of the Course: Petroleum Refining Engineering

Course Code: TPE-505

Time: Three Hours

MM: 100

Note:

- All questions are compulsory.
- Answer any two sub questions in each main question.
- Total marks for each main question is **twenty**
- Each Question carries **10 marks**

Q1		COs
a.	What are the different physical involved in the refining of crude oil refining? Explain with flow sheet diagram for any one physical process with process description	CO1, CO2
b.	What do you mean by quality control of the petroleum products? Hence Define the following: - Delayed coking - Cloud point - Visbreaking - Pour point - Fire point - Freezing point - Cetane number - Electrostatic separation - Demulsifier - Azeotropic Distillation	CO1, CO2
c.	Discuss the elementary composition of crude oil? Draw the flow sheet diagram of Vacuum distillation unit (VDU) and discuss its process description step wise?	CO3
Q2		
a.	Discuss continuous distillation classification based on - Nature of feed - Number of product stream - Types of column internals - Extra feed exits	CO3, CO4
b.	Explain the different main components of the distillation column. Discuss role of plate and trays and packers in the distillation column	CO1, CO4
c.	Design the flow sheet of aromatic solvent extraction unit? What are the purposes of solvent extraction? What are the widely used extraction solvents?	CO1, CO5
Q3		
a.	Explain the general steps of solvent dewaxing? Discuss Vis breaking with the flow sheet diagram and illustrate the process description.	CO1
b.	Explain the role of complete and incomplete wetting of the catalyst in trickle bed	CO1, CO2

	reactor? Discuss hydro treating with the neat diagram of trickle bed reactor?	
c.	Explain the different catalytic processes? How cracking is catalyzed? What are the other reactions during the catalytic cracking, explain one of them?	CO2, CO5
Q4		
a.	Design the hydro cracking flow sheet scheme unit and explain their process description?	CO1, CO5
b.	Define the process hydro treating and? Why we do hydro treating? Illustrate catalytic hydrodesulphurization in details with the help of trickle bed reactor diagram.	CO4, CO5
c.	Why we do alkylation to the residues obtained? Explain the sulfuric acid alkylation with the suitable flow sheet diagram and its process description?	CO1, CO6
Q5		
a.	What do you mean by reformates of refining? What are the catalytic reforming reactors? Draw and explain the process description of the flow sheet scheme of catalytic reforming.	CO4, CO6
b.	Discuss fluid coking and flexicoking? Why flexicoking is better than fluid coking?	CO4, CO5
c.	Explain catalyst deactivation and need of catalyst deactivation? What are the different reactors used for catalytic hydrogenation? Explain them with the labeled diagram.	CO2, CO3